

Upstream Scenario Modelling Framework

User Guide

Guidance for preparing inputs, configuring scenarios, running the model and reviewing outputs.



Prepare Inputs



Configure



Validate



Run



Review

Transparent • Repeatable • User-controlled scenario analysis

Contents

What this guide covers, in the order you will use the modeller.

1

Getting Around the Modeller

The three principal screens and how they fit together.

2

Home – Control the Modelling Workflow

Refresh inputs, run the engine and review status from one place.

3

Configure a Scenario

Include cases, select operations and enter parameters with validation.

4

Review and Compare Scenario Results

Select scenarios and metrics, then compare profiles side by side.

5

Financial Entity to Modelling Case Translation

How financial entities are mapped to modelling cases.

6

Add a Manual Modelling Case

Use the standardised template to add a case to the input set.

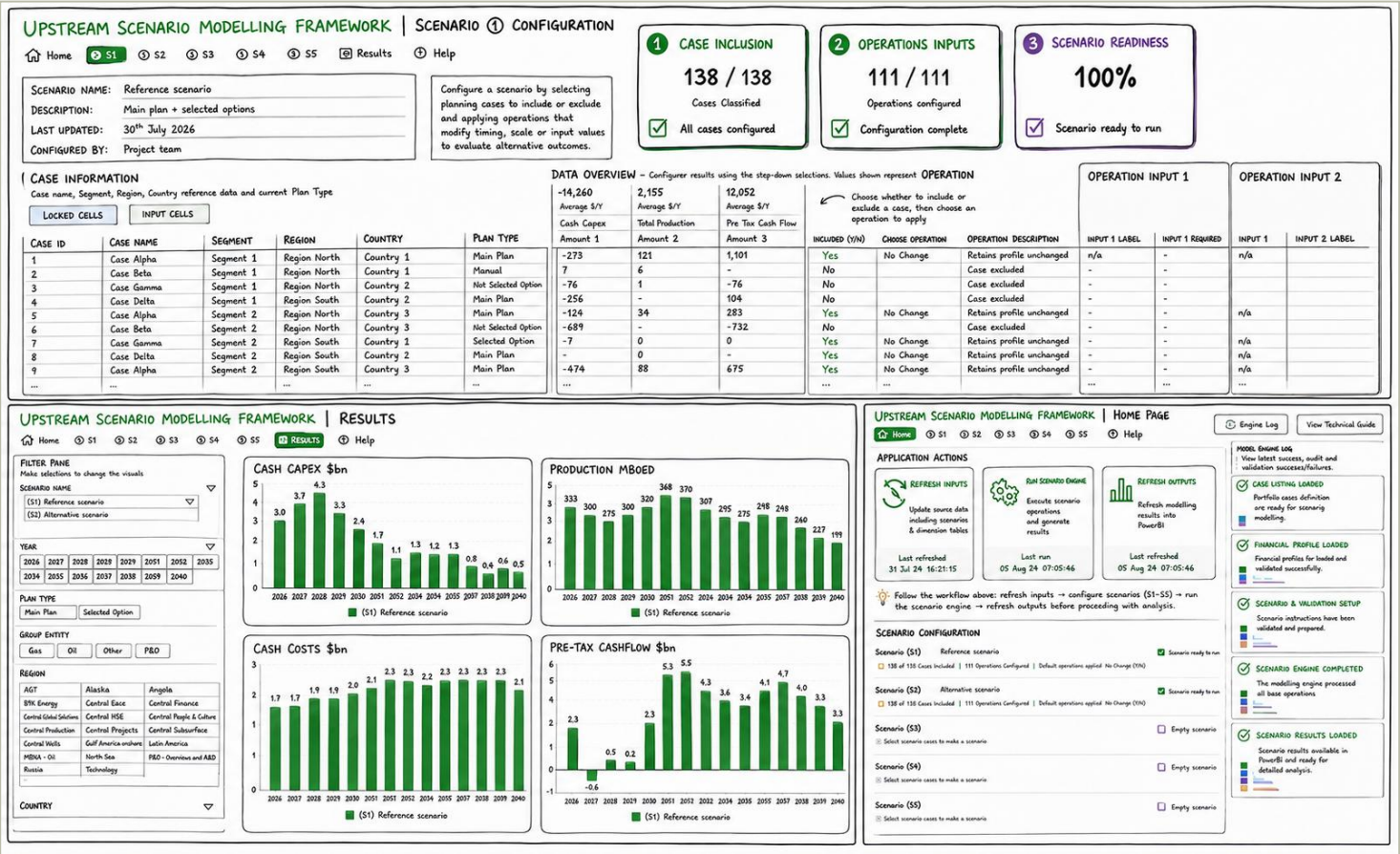
7

Operations Library

The operations available when configuring a scenario.

Getting Around the Modeller

The application is organised around a simple workflow from model control, through scenario configuration, to results analysis.



- 1 HOME**
Control the modelling workflow.
 - Refresh model inputs
 - Run the scenario engine
 - Refresh scenario outputs
 - Review validation and model status
- 2 SCENARIO CONFIGURATION**
Define what each scenario should do.
 - Include or exclude modelling cases
 - Select scenario operations
 - Enter operation parameters
 - Review scenario readiness before running

- 3 RESULTS**
Review and compare scenario outcomes.
 - Select scenarios for comparison
 - Filter the portfolio
 - Review key financial and production metrics
 - Analyse impacts across the planning horizon

HOME → CONFIGURE SCENARIOS → RUN ENGINE → REFRESH OUTPUTS → REVIEW RESULTS

Five scenario configuration pages (S1-S5) are available and use the same structure and validation controls.

Home – Control the Modelling Workflow

The Home page provides the central control point for refreshing data, running the scenario engine and monitoring model status.

Upstream Scenario Modelling Framework | Home Page

Home

1

S1

2

S2

3

S3

4

S4

5

S5

Results

Help

Application Actions

Refresh Inputs

Update source data including model cases and financial profiles

Last refreshed: 18 Aug 26 08:44:07

Run Scenario Engine

Execute scenario operations and generate modelling outcomes

Last run: 18 Aug 26 09:14:45

Refresh Outputs

Refresh modelling results into Pivottables and dashboard visuals

Last refreshed: 18 Aug 26 09:15:00

Follow the workflow above: refresh inputs → configure scenarios (S1-S5) → run the scenario engine → refresh outputs **before** proceeding with analysis.

Scenario Configuration

Scenario (S1)

Reference case

12 (of 12) Cases Included | 12 Operations Configured | Distinct operations used: No Change [12]

Scenario ready to run

Scenario (S2)

Delay case

12 (of 12) Cases Included | 12 Operations Configured | Distinct operations used: Delay [12]

Scenario ready to run

Scenario (S3)

Exclude some cases

9 (of 12) Cases Included | 9 Operations Configured | Distinct operations used: No Change [9]

Scenario ready to run

Scenario (S4)

Add a large capex and opex amount

2 (of 12) Cases Included | 2 Operations Configured | Distinct operations used: CAPEX Change \$m [1], OPEX Change \$m [1]

Scenario ready to run

Scenario (S5)

Empty scenario

Select some cases to make a scenario

Empty scenario

Model Engine Log

Run duration: 14.52 seconds

Case listing loaded and validated successfully.

Portfolio case definitions are ready for scenario modelling.

12 valid rows retained from 12 imported.
0 invalid records removed.
0 rows contain incomplete hierarchy attributes.
12 unique case IDs and 12 unique case names validated.

Financial profiles loaded and validated successfully.

Financial profiles for 12 cases are ready for scenario modelling across the 2026–2040 planning horizon.

12 valid rows retained from 12 imported.
0 invalid records removed.
0 rows contain incomplete hierarchy attributes.
12 unique case IDs and 12 unique case names validated.

Scenario instructions prepared successfully.

Scenario instructions have been validated and prepared for execution across all configured scenarios.

60 instruction rows processed.
0 operations on excluded cases cleared.
0 blank or invalid operations defaulted to NoChange.
0 stale parameter sets reset.

Scenario engine completed successfully.

The modelling engine processed 35 case operations and generated updated scenario results.

4 scenarios completed; 1 skipped.
35 case operations processed.
9,490 rows generated across 12 cases.
0 cases have incomplete hierarchy attributes.

1

REFRESH INPUTS
Refresh the datasets used by the Excel application before configuring or running scenarios.

2

CONFIGURE SCENARIOS
Use the S1–S5 navigation buttons to move to the five scenario configuration pages.

3

RUN ENGINE
Execute the configured scenarios through the modelling engine once inputs and instructions are ready.

4

REFRESH OUTPUTS
Refresh the scenario-results dataset and analytical outputs after a successful engine run.

5

REVIEW MODEL STATUS
Use the validation and status information to confirm input readiness, execution and any issues to address.

REFRESH INPUTS → CONFIGURE → RUN ENGINE → REFRESH OUTPUTS → REVIEW

TRANSPARENCY

Model health and execution messages make processing status visible to the user.

USER CONTROL

Scenario execution and output refresh remain explicit user actions rather than automatic background processes.

GOOD PRACTICE – before running the model

- Confirm the required inputs have refreshed successfully.
- Review scenario configuration and validation messages.
- Resolve any reported errors before relying on refreshed outputs.

Configure a Scenario

Each scenario page allows the user to select modelling cases, apply controlled operations and define the parameters used by the scenario engine.

Upstream Scenario Modelling Framework | Scenario 1 Configuration

[Home](#) [S1](#) [S2](#) [S3](#) [S4](#) [S5](#) [Results](#) [Help](#)

Scenario NameReference case

Description

Last Updated

Configured By

Configure a scenario by selecting planning cases to **include** or **exclude** and applying operations that modify **timing**, **scale** or **input values** to evaluate alternative outcomes.

Case Information

Case name, Segment, Region, Country reference data and current Plan Type

Locked Cells

Input Cells

Case ID	Case Name	Segment	Region	Country	Plan Type	Amount 1	Amount 2	Amount 3	Included (y/n)	Choose Operation	Operation Description	Input 1 Label	Input 1 Required	Input 1
1	Project Ironwood	G&LC	Frontier Region	Eastern Province	Main Plan	-129	-129	-129	Yes	No Change	Retains profile unchanged	n/a	-	
2	Project Northstar	G&LC	Frontier Region	Eastern Province	Main Plan	-122	-122	-122	Yes	No Change	Retains profile unchanged	n/a	-	
3	Project Silver Peak	P&O	Falcon Territories	Falcon Region	Main Plan	-947	-947	-947	Yes	No Change	Retains profile unchanged	n/a	-	
4	Project Granite Basin	G&LC	Falcon Operations	Falcon Block	Main Plan	-47	-47	-47	Yes	No Change	Retains profile unchanged	n/a	-	
5	Project Copper Bay	P&O	Coastal Republic	Coastal Republic	Main Plan	-10	-10	-10	Yes	No Change	Retains profile unchanged	n/a	-	
6	Project Redcliff	G&LC	Pacific Region	Southern Coast	Main Plan	-112	-112	-112	Yes	No Change	Retains profile unchanged	n/a	-	
7	Project Highland Crest	G&LC	Meridian Region	Meridian Region	Main Plan	-13	-13	-13	Yes	No Change	Retains profile unchanged	n/a	-	
8	Project Blue Horizon	P&O	Central Oil Region	Desert Province	Main Plan	-588	-588	-588	Yes	No Change	Retains profile unchanged	n/a	-	
9	Project Cedar Point	P&O	Northern Shelf	Northern Shelf West	Manual	-460	-460	-460	Yes	No Change	Retains profile unchanged	n/a	-	
10	Project Stonehaven	P&O	Northern Shelf	Northern Shelf West	Main Plan	-26	-26	-26	Yes	No Change	Retains profile unchanged	n/a	-	
11	Project Raven Rock	G&LC	Island Region	Island Region	Main Plan	-164	-164	-164	Yes	No Change	Retains profile unchanged	n/a	-	
12	Project Eagle Pass	P&O	Western Hemisphere	Southern Republic	Manual	-460	-460	-460	Yes	No Change	Retains profile unchanged	n/a	-	
13	No case	-	-	-	-	-	-	-	-	-	-	-	-	-

1 Case Inclusion

12 / 12

Cases Classified

All cases configured

2 Operations Inputs

12 / 12

Operations configured

Configuration complete

3 Scenario Readiness

100%

Scenario ready to run

Data Overview

Configure results using the drop-down selections. Values shown represent the

Operation

Choose whether to include or exclude a case, then choose an operation to apply

-3,077

Average 5Y

Cash Capex

-3,077

Average 5Y

Cash Capex

-3,077

Average 5Y

Cash Capex

Operation input 1

INCLUDE CASE → SELECT OPERATION → ENTER PARAMETERS → CHECK VALIDATION → RUN

Five scenario pages (S1-S5) are available, each using the same configuration structure so alternative portfolio scenarios can be prepared and compared consistently.

- 1

SCENARIO DETAILS
Name and describe the scenario so its purpose and assumptions are clear when results are reviewed and compared.
- 2

SELECT CASES
Use the Include control to determine which modelling cases participate. Cases not included are excluded from scenario processing.
- 3

CHOOSE AN OPERATION
Select a modelling operation from the controlled drop-down list. An included case with no operation selected is treated as "No Change".
- 4

ENTER PARAMETERS
Complete the Input fields required by the operation – their meaning changes with the operation selected, and embedded guidance identifies each expected parameter. Data validation constrains Inputs 1-3 where appropriate.
- 5

REVIEW CASE CONTEXT
Use Group Entity, Segment, Region, Country and Plan Type to confirm the intended case and to give context when analysing the scenario.
- 6

CHECK VALIDATION
Review scenario validation and readiness before running. Invalid instructions prevent refreshed results from being published, and errors are surfaced to the user.

GUIDED CONFIGURATION

Controlled operation lists, configurable input validation and operation-specific guidance prevent invalid instructions at the point of entry.

LAYERED VALIDATION

Excel provides preventative input controls during configuration; the modelling engine applies business-rule validation during execution.

HUMAN OVERSIGHT

The user decides which cases to include, which operations to apply and which assumptions to enter. The application executes, it does not choose the scenario.

Review and Compare Scenario Results

The Results page provides an interactive view of scenario outcomes, allowing users to compare financial and production impacts across the portfolio and planning horizon.



- 1 SELECT SCENARIOS**
Use the Scenario selector to choose the scenarios to review and compare. Names entered during configuration carry through to the results, relating outputs back to the assumptions being tested.
- 2 FILTER THE PORTFOLIO**
Use the portfolio filters to focus on the required part of the portfolio, exploring results across the organisational, geographic, planning and case classifications available in the model.
- 3 SELECT THE METRIC**
Choose the financial or production metric required. The interface gives a consistent analytical view across supported metrics, so different scenario impacts are examined the same way.

SELECT SCENARIOS → FILTER PORTFOLIO → SELECT METRIC → REVIEW PROFILE → COMPARE

- 4 REVIEW THE TIME PROFILE**
Use the results table and chart to see how the selected metric changes across the planning horizon, identifying the magnitude and the timing of impacts.

- 5 COMPARE SCENARIO OUTCOMES**
Compare alternative scenario profiles to understand the effect of different case selections, operations and assumptions, using the visual and tabular views together.

- 6 REFINE THE ANALYSIS**
Adjust scenario, portfolio and metric selections to investigate the drivers behind a result. The interface supports iterative exploration rather than a single prescribed answer.

TRACEABILITY

Scenario identity and modelling context are retained through to the results, so outputs can be related back to the configuration that generated them.

CONSISTENT ANALYSIS

Common metrics, filters and visual structures allow alternative scenarios to be reviewed within a consistent analytical framework.

DECISION SUPPORT

The application presents the calculated consequences of user-defined scenarios. Interpretation, wider business factors and decisions remain with the user.

Scenario outputs should be reviewed alongside the assumptions and operations used to create them. A modelled result is the consequence of the configured scenario – not a recommendation or prediction of the preferred portfolio outcome.

The translation template provides a controlled way to define how source Financial Entities are grouped and presented as modelling cases within the Scenario Modeller.

[illegible][illegible]

4 PROVIDE A CASE NAME

Enter a clear, business-readable Case Name – the label users see when configuring scenarios and reviewing outputs, without needing to interpret entity identifiers.

The required modelling representation is maintained separately from the governed Financial Entity structure.

5 COMPLETE THE CLASSIFICATION

Maintain the supporting attributes required by the Scenario Modeller – Group Entity, Segment, Region, Country and Plan Type – for context, filtering and analysis.

Overrides handle exceptions without changing the underlying entity or embedding special cases in the engine.

6 REVIEW THE EFFECTIVE MAPPING

Before refreshing, confirm each Financial Entity is represented by the intended case, paying particular attention to records using the override.

Entities, cases and explicit overrides stay visible and inspectable rather than hidden in calculation logic.

7 REFRESH MODEL INPUTS Once reviewed and saved, return to the Home page and refresh model inputs so the new mapping determines the cases and attributes available.

Source identity, modelling structure and scenario calculations stay separate, so representation changes need no engine change.

1 REVIEW FINANCIAL ENTITIES

Confirm the source records to be represented in the application. The Financial Entity stays the governed source reference; the translation layer decides how it is grouped and presented.

2 ASSIGN A MODELLING CASE

Assign each relevant Financial Entity to a Modelling Case. Several entities can map to a common case where they should be treated together for scenario analysis.

2 USE THE OVERRIDE WHERE REQUIRED

Where the standard assignment does not give the required business representation, use the controlled override; leave it blank where the standard mapping applies. The exception stays visible and the entity reference is retained.

Changes to the translation template affect how source profiles are grouped and presented in the Scenario Modeller. Review standard mappings and overrides carefully before refreshing model inputs.

Add a Manual Modelling Case

The standardised manual input template allows representative profiles that are not available through the governed source-data pipeline to be incorporated into the Scenario Modeller using the same controlled structure as system-sourced cases.

bp

LTP Manual Data Feed

Update the case name, reference data, and financial profiles

Case name

Project Blackwater

<< Input

All OK

Segment

P&O

<< Select

All OK

Entity

Oil

<< Select

All OK

Region

North Sea

<< Select

All OK

Country

North Sea - UK

<< Select

All OK

Plan Type

Manual

<< Select

All OK

This is a very good project, a very good one indeed

ID	Metric	Notes	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
0000002020	Total Production (mboed)		30	40	40	40	40	30	30	30	30	30	20	20	20	20	20
0000000310	Gas Production (mmscf/d)		0	10	10	10	10	0	0	0	0	0					
0000000080	Oil Production (mb/d)		30	40	40	40	40	30	30	30	30	30	20	20	20	20	20
0000000190	NGL Production (mb/d)		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000000001	Liquids Production (mboed)	Calculated: Oil + NGL	30	40	40	40	40	30	30	30	30	30	20	20	20	20	20
0000003770	Revenue Generating Total Prod (mboed)		40	40	40	40	40	30	30	30	30	30	20	20	20	20	20
0000014411	Equity Accounted Total Production (mboed)		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000000002	Rev Gen Prod less EQ Prod (mboed)	Calculated: Rev gen + Equity Accounted	40	40	40	40	40	30	30	30	30	30	20	20	20	20	20
0000000003	Gross Margin		1340	1180	1240	1200	1170	1130	1100	1040	1020	990	930	870	820	780	750
0000000004	Total Cash Costs		-270	-270	-250	-250	-250	-250	-250	-250	-250	-250	-250	-260	-260	-260	-260
0000000005	Production Costs		-180	-190	-190	-180	-180	-190	-190	-190	-190	-190	-200	-200	-200	-200	-210
0000000006	DD&A Charge		-610	-690	-700	-660	-630	-590	-560	-510	-490	-470	-430	-390	-370	-340	-320
0000000007	Related Company Income		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0000000009	Underlying RCOP		390	230	300	300	310	300	300	290	290	280	260	230	200	180	170
	Headline RCOP		n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
0000000008	EBITDA (Underlying)		1070	910	990	950	920	880	850	790	770	740	680	610	560	520	490
	Tax Charge	Calculated: RCOP underlying x Tax rate	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
	Post-Tax RCOP (Underlying)	Calculated: RCOP underlying + Tax charge	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a

1 **USE THE STANDARD TEMPLATE**
Use the controlled manual input template where a required case or adjustment is not available from the standard source-data pipeline. Do not change the template structure, worksheet layout or controlled fields.

2 **COMPLETE THE CASE METADATA**
Provide the information that identifies and classifies the case – Case Name plus Group Entity, Segment, Region, Country and Plan Type. Use clear business-readable descriptions, as these identify the case in the Scenario Modeller.

3 **ENTER THE PROFILE DATA**
Populate the required financial and production profiles across the relevant planning years using the standard profile structure. Keep the expected metric names, year structure and units so the profile is processed consistently.

Manual inputs should be used where a required profile cannot appropriately be represented through the standard source-data pipeline. Users remain responsible for the completeness, accuracy and business appropriateness of manually supplied assumptions.

COMPLETE METADATA → ENTER PROFILE → REVIEW → SAVE TO CONTROLLED LOCATION → REFRESH INPUTS → CONFIGURE SCENARIO

4 **REVIEW THE INPUT**

- required metadata is complete;
- the Case Name clearly identifies the case;
- values sit against the correct metrics and years;
- the template structure is unchanged;
- only the required information is included.

5 **SAVE TO THE CONTROLLED LOCATION**

Save the completed workbook to the designated controlled manual-feed location. The ingestion process picks up supported workbooks from there and incorporates their profile and metadata.

6 **REFRESH MODEL INPUTS**

Return to the Home page and refresh model inputs. The manual case then joins the available modelling population and can be selected and configured alongside other cases.

7 **CONFIGURE AS A NORMAL CASE**

Once ingested, the manual profile follows the common modelling structure – it can be included in scenarios, given supported operations and analysed through the same results workflow.

STANDARDISED INPUT

A common template structure lets manually supplied profiles enter the framework consistently, without bespoke calculation logic.

DATA MINIMISATION

Provide only the metadata and profile information required for the defined modelling purpose.

TRACEABILITY

Manual cases keep their own identity and classification, so their presence can be recognised and followed through the workflow.

CONSISTENT TREATMENT

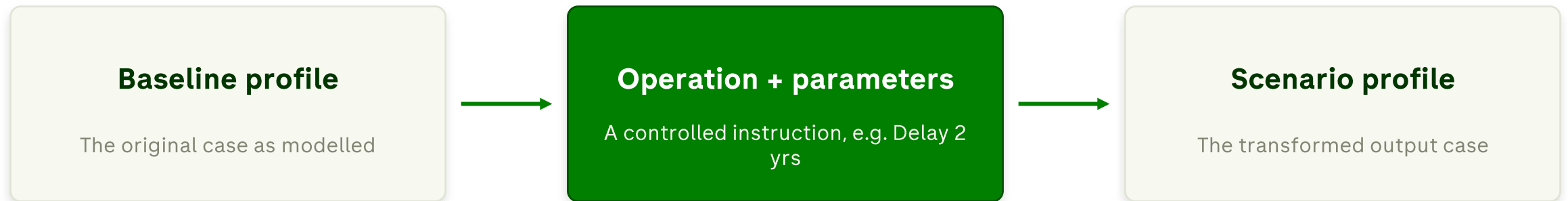
Once incorporated, manual cases use the same scenario configuration, operation and analytical framework as other cases.

Operations Library

User Guide

An operation is a controlled modelling instruction applied to a case profile.

It changes timing, scale, production, cost, CAPEX or transaction assumptions – without manually rebuilding the source profile.



Same source data, applied consistently – no manual profile rebuild.

Why operations matter

CONSISTENT, REPEATABLE, AUDITABLE

Operations replace ad-hoc spreadsheet edits with a controlled, transparent way to build scenarios.

01

Faster scenario creation

Build cases in a few inputs rather than hours of rework.

02

Controlled assumptions

Every change is explicit, parameter-driven and visible.

03

Consistent metric treatment

Impacts flow through the model the same way each time.

04

Easier comparison

Cases and scenarios stay directly comparable.

05

Less manual manipulation

Fewer spreadsheet errors and version-control risks.

Operation library overview

TWELVE OPERATIONS, FOUR CATEGORIES

Default

1

Baseline

- No Change

Timing

4

When it happens

- Delay
- Accelerate
- Truncate Before
- Truncate After

Scale

2

How much of it

- Farm In
- Farm Down

Input metric

5

The value itself

- Production Change %
- OPEX Change %
- CAPEX Change %
- CAPEX Change \$m
- OPEX Change \$m



Timing operations

SHIFT OR TRIM THE PROFILE ALONG THE YEAR AXIS

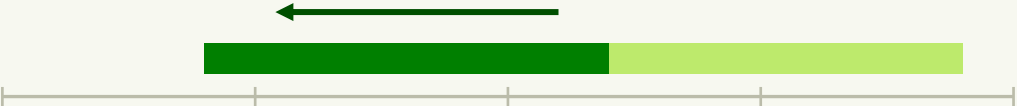
Delay

Moves the profile later. Optional delay penalty and penalty year.



Accelerate

Moves the profile earlier. Optional funding requirement and funding year.



Truncate Before

Removes all values before the selected year.



Truncate After

Removes all values after the selected year.



Scale and transaction operations

RESIZE THE PROFILE – AND CAPTURE THE DEAL

Farm In

Increase



Increases the selected profile to reflect a larger working interest.

Can include an acquisition payment (\$m).

Farm Down

Dilution



Reduces the selected profile to reflect a smaller working interest.

Can include disposal proceeds (\$m).

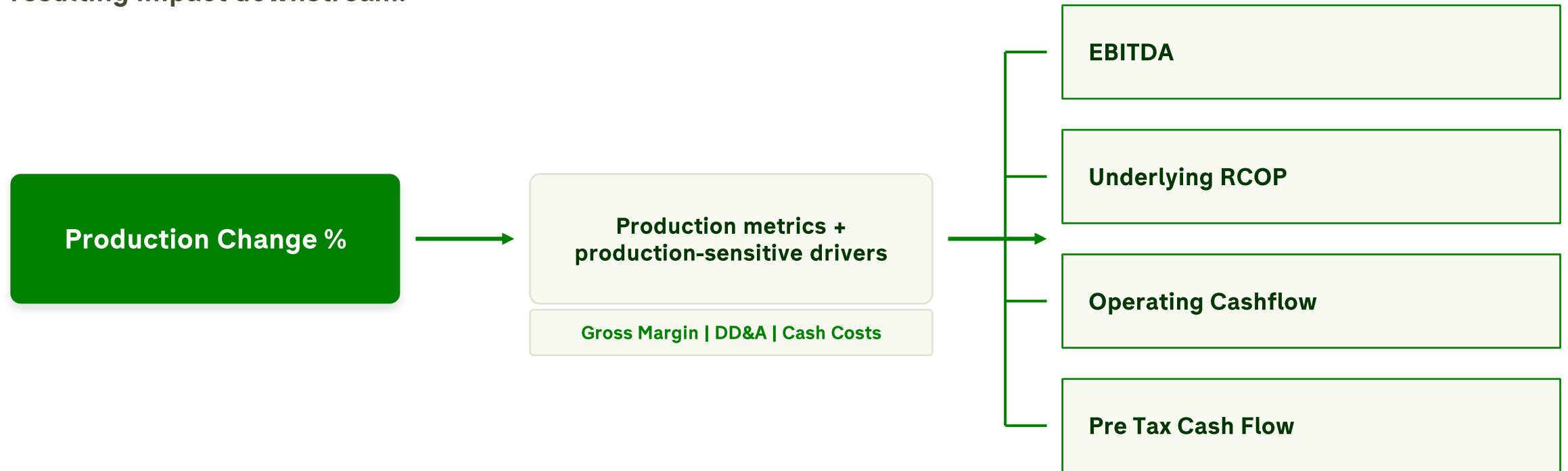
Where proceeds land

Acquisition payments and disposal proceeds flow into **Cash Flow from Sale of Operations** and, in turn, **Pre Tax Cash Flow**.

Production adjustment

PRODUCTION CHANGE %

Production Change % scales production metrics and production-sensitive financial drivers, **then propagates the resulting impact downstream.**



Note Composite metrics do not always move by the same percentage – they include other fixed or unaffected items.



Cash Cost and Cash Capex adjustments

PROPORTIONAL OR SPECIFIC \$M CHANGES

Cash Cost Change %

Adjusts **Total Cash Costs** proportionally

Cash Capex Change %

Adjusts **Cash Capex** proportionally

Cash Cost Change \$m

Specific annual **cost additions / reductions**

Cash Capex Change \$m

Specific annual **funding additions / reductions**

Worked example – Cash Capex Change \$m

Specific ± \$m adjustments applied to Cash Capex by year

Year	Cash Capex base (\$m)	Change (\$m)	Adjusted (\$m)
2027	120	+15	135
2028	90	-10	80

Reductions carry a minus sign; additions a plus. Each year is set explicitly.



Parameter design

One generic Input 1–5 structure for every operation. The same five-input structure supports different operation types while keeping the Excel interface consistent.

Operation	Category	Description	Input 1 Label	Req	Input 2 Label	Req	Input 3 Label	Req	Input 4 Label	Req	Input 5 Label	Req
No Change	Default	Retains profile unchanged	n/a	-	n/a	-	n/a	-	n/a	-	n/a	-
Delay	Timing	Moves profile later	Years to delay (Yrs)	Yes	Delay penalty (\$m)	Optional	Penalty year	Optional	n/a	-	n/a	-
Accelerate	Timing	Moves profile earlier	Years to accelerate	Yes	Funding required (\$m)	Optional	Funding year	Optional	n/a	-	n/a	-
Farm In (Increase)	Scale	Increases selected profile	Increase (%)	Yes	Effective start year	Optional	Effective end year	Optional	Acquisition payment (\$m)	Optional	Payment year	Optional
Farm Down (Dilution)	Scale	Reduces selected profile	Dilution percentage (%)	Yes	Effective start year	Optional	Effective end year	Optional	Disposal proceeds	Optional	Proceeds year	Optional
Truncate Before	Timing	Removes values before year	Retain profile from year	Yes	n/a	-	n/a	-	n/a	-	n/a	-
Truncate After	Timing	Removes values after year	Retain profile until year	Yes	n/a	-	n/a	-	n/a	-	n/a	-
Production Change %	Input metric	Adjusts production by a %	Percentage change	Yes	Start year	Optional	End year	Optional	n/a	-	n/a	-
OPEX Change %	Input metric	Adjusts OPEX by a %	Percentage change	Yes	Start year	Optional	End year	Optional	n/a	-	n/a	-
CAPEX Change %	Input metric	Adjusts CAPEX by a %	Percentage change	Yes	Start year	Optional	End year	Optional	n/a	-	n/a	-
CAPEX Change \$m	Input metric	Adjusts CAPEX by \$m	Change Yr1 (± \$m)	Yes	Year 1	Yes	Change Yr2 (± \$m)	Optional	Year 2	Optional	n/a	-
OPEX Change \$m	Input metric	Adjusts OPEX by \$m	Change Yr1 (± \$m)	Yes	Year 1	Yes	Change Yr2 (± \$m)	Optional	Year 2	Optional	n/a	-